

KAMAL MAHER

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Research interests: AI agents · interpretability · graph ML · biological foundation models

EDUCATION

2020 – 2025 | **MIT**, PhD IN COMPUTATIONAL BIOLOGY
Advisor: Xiao Wang

2014 – 2018 | **Cornell University**, BS IN NEUROSCIENCE

EXPERIENCE

2025 – 2026 | **SPAR**, RESEARCH FELLOW

- Built an agentic framework that iteratively probes and explains learned representations in large language models, improving feature explanation accuracy by 7.7% over existing methods. [1]
- Demonstrated that a widely-used mechanistic interpretability method is incentivized to learn unfaithful explanations of neural network computation. [2]

2021 – 2025 | **MIT**, GRADUATE RESEARCHER

- Developed a spectral graph framework linking tissue regions and cell–cell interactions via harmonic analysis of spatial transcriptomics. [3]
- Applied the framework to a 1.09M cell mouse brain atlas, revealing transcriptional gradients and regions absent from established anatomical references. [4]

2024 | **Genentech**, AVIV REGEV LAB INTERN

- Applied graph signal processing methods to cancer spatial transcriptomics data, identifying regions and cell–cell interactions more efficiently than existing GNN-based approaches.

PUBLICATIONS (*equal contribution)

[1] *Multi-shot AutoInterp: Agents Can Explain Complex Features By Refining Explanations.*

K. Maher*, S.E. Schrader*, K. Ayonrinde

ICLR 2026 Workshop on Unifying Concept Representation Learning. 2026

ICLR 2026 Workshop — From Human Cognition to AI Reasoning. 2026

[2] *Cross-Layer Transcoders are Incentivized to Learn Unfaithful Circuits.*

G. Lange, R. Goldstein, K. Dearstyne, **K. Maher**

LessWrong. 2026

[3] *Harmonic Representations of Regions and Interactions in Spatial Transcriptomics.*

K. Maher, X. Wang

bioRxiv. 2024

[4] *Spatial Atlas of the Mouse Central Nervous System at Molecular Resolution.*

H. Shi*, Y. He*, Y. Zhou*, J. Huang, **K. Maher**, B. Wang, Z. Tang, S. Luo, P. Tan, M. Wu, Z. Lin, J. Ren, Y. Thapa, X. Tang, K.Y. Chan, B.E. Deverman, H. Shen, A. Liu, J. Liu, X. Wang

Nature. 2023

SKILLS

ML / AI: representation learning · mechanistic interpretability · graph ML · agentic workflows

Tools: PyTorch · Git · $\text{\LaTeX}$ · Cloud (Runpod, Vast) · HPC (SLURM, UGE)

Biology: spatial transcriptomics · tissue biology · immunology · neuroscience